e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:07/Issue:05/May-2025 Impact Factor- 8.187 www.irjmets.com

AGRIVERSEAI – SMART AGRICULTURAL WEB APPLICATION FOR KARNATAKA PEOPLE

Prof. Sagar Birje*1, Mr. Kiran B Nandani*2, Mr. Suprit K.L.*3, Mr. Sunil Naik*4, Mr. Anish Inamadar*5

*1Assistant Professor & HOD, Artificial Intelligence & Data Science, Angadi Institute Of Technology & Management, Belagavi, Karnataka, India.

*2,3,4,5UG Student, Dept. Artificial Intelligence & Data Science, Angadi Institute Of Technology & Management, Belagavi, Karnataka, India.

DOI : https://www.doi.org/10.56726/IRJMETS78194

ABSTRACT

AgriVerseAI is a voice-activated, bilingual smart agriculture assistant designed for Kannada-speaking farmers in Karnataka. It uses advanced speech technologies and natural language processing (NLP) to overcome challenges like low digital literacy and language barriers. The assistant provides real-time support for crop advisory, soil analysis, disease diagnosis, market insights, insurance, and government schemes. With its voice-first interface, even farmers with limited literacy can access localized, actionable information effortlessly. The system integrates speech-to-text, language detection, knowledge bases, and text-to-speech to ensure accurate, context-aware assistance. AgriVerseAI offers a scalable solution to strengthen agricultural outreach and bridge information gaps in rural communities.

Keywords: Agriverseai, Smart Agriculture, Voice Assistant, NLP, Kannada, Rural Technology.

I. INTRODUCTION

Agriculture is a vital part of Karnataka's economy, supporting a significant segment of its population. Farmers in the region encounter various challenges, including diverse climate conditions, different soil types, and limited access to timely, localized information due to low literacy rates. These obstacles impede informed decision-making and hinder the adoption of modern farming practices that could enhance productivity and income.

To address these issues, we have created AgriVerseAI—an AI-driven voice assistant that supports both Kannada and English, making it accessible to a broad audience in rural areas. With a conversational voice interface, it reduces the need for written input and improves usability for individuals with limited formal education.

At its foundation, AgriVerseAI employs a robust NLP engine that provides intelligent, real-time guidance on critical farming activities, including:

- Selecting crops suited to local conditions
- Determining optimal planting times for improved yields
- Implementing best practices for harvesting and post-harvest management
- Conducting soil tests and managing nutrients
- Identifying and treating common plant diseases and pests
- In addition to agronomic advice, the tool offers economic and administrative insights such as current market prices, selling strategies, and information about government policies and crop insurance. These features empower farmers to make informed decisions, secure better returns, and reduce financial risks.

By leveraging AI and multilingual voice technology, AgriVerseAI aims to foster sustainable farming practices and support the socio-economic development of Karnataka's agricultural communities. This innovative approach illustrates how smart, inclusive technology can transform traditional agriculture into a more efficient and resilient sector.

II. METHODOLOGY

The development of AgriVerseAI employs an interdisciplinary approach that integrates artificial intelligence, NLP, and voice technologies to create a user-friendly bilingual assistant tailored for Karnataka's farming

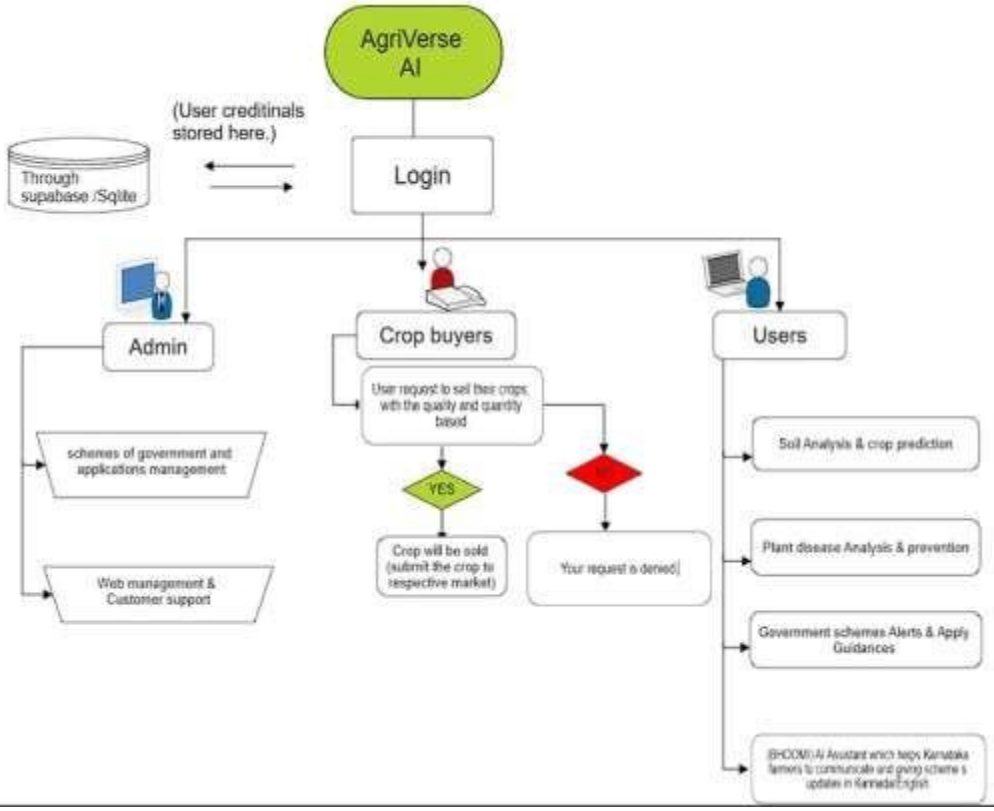
e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:07/Issue:05/May-2025 Impact Factor- 8.187 www.irjmets.com

community. This section details the system architecture, data collection methods, conversation design, AI integration, and validation techniques used to ensure a robust and intuitive solution.

AgriVerseAI is centred around an intelligent NLP-driven chatbot engine composed of several integrated modules that process voice inputs, interpret text, retrieve relevant knowledge, and generate spoken outputs. Key components include:

- **Speech-to-Text Module**: Converts spoken queries in Kannada or English into written form for further processing.
- **NLP Chatbot Core**: Manages intent recognition, entity extraction, and contextual understanding to provide meaningful responses.
- **Agricultural Knowledge Base**: Contains structured data on crops, soil types, diseases, market rates, and policy information.
- **Advisory Generation Engine**: Utilizes AI models and logic rules to offer personalized recommendations for crop care and disease diagnosis.

III. SYSTEM ARCHITECTURE



AgriVerseAI employs advanced speech-to-text and text-to-speech technologies to support both Kannada and English, ensuring accurate voice recognition across regional accents. Its bilingual NLP engine includes intent detection, named entity recognition, and context tracking to facilitate smooth, multi-turn conversations. Automatic language detection allows users to switch between languages seamlessly within a single session.

The system relies on a comprehensive, curated agricultural database sourced from Karnataka's crop records, soil maps, market price lists, and government programs. Its voice interface is designed for simplicity, using everyday language, memory retention, and error recovery to accommodate users with low literacy.

Crop disease diagnosis combines rule-based symptom matching with machine learning models to provide accurate and region-specific advice. Field trials, expert evaluations, and user feedback were conducted to validate AgriVerseAI, ensuring practical utility and acceptance among rural farmers.

e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:07/Issue:05/May-2025 Impact Factor- 8.187 www.irjmets.com

IV. FUNCTIONALITY & MODULES

Crop Advisory

This feature provides tailored crop recommendations based on local climate, seasonal patterns, and soil characteristics. Farmers receive guidance on optimal planting schedules, irrigation methods, and fertilizer application, helping them maximize yield and resource efficiency.

Soil Analysis

Using a question-and-answer format, the system collects information about soil characteristics such as colour, texture, moisture retention, and cracking patterns. Based on this input, it identifies the likely soil type and suggests appropriate treatments like liming, composting, or nutrient enrichment. This enables farmers without access to laboratory testing to effectively assess soil health.

Disease Diagnosis

Farmers can describe symptoms such as leaf spots, discoloration, or wilting through voice input. The system compares these descriptions with a database of known diseases and provides both organic and chemical treatment options. This helps farmers respond promptly and minimize losses.

Market Price and Selling Strategy

This module integrates live market price data to provide up-to-date pricing information. Users receive guidance on when and where to sell their produce for maximum profit. It also suggests optimal harvest times aligned with demand cycles.

Government Schemes and Insurance

This section informs farmers about state and national-level initiatives, including subsidies, crop insurance, and financial assistance. It explains eligibility requirements, application processes, and benefits in simple terms, facilitating easier access to institutional support.

Together, these modules enable AgriVerseAI to deliver comprehensive and actionable agricultural advice, reducing information gaps and empowering farmers to make informed decisions throughout the growing season.

V. NEW IMPLIMENTATIONS

AgriVerseAI is developed using **React.js** for the web interface and **Flutter** for mobile applications, ensuring a responsive and visually appealing experience across devices. The backend is powered by **Python**, utilizing **Flask** and **Fast API** to manage user requests and connect to **OpenAI GPT-3/Gemini 2.0/2.5 Pro APIs** for bilingual NLP capabilities.

For voice input, **Google Speech-to-Text** handles English while a custom-trained **Mozilla Deep Speech** model supports Kannada, accurately capturing regional dialects. Language

detection is managed using **Lang detect** and **fast Text**, routing inputs to the appropriate NLP pipeline. Responses are converted into natural-sounding audio using **Google Text-to-Speech** and **Coqui.ai**, enhancing user interaction with clear, localized voice output.

The system is hosted on platforms like **Render**, **Heroku**, and **Digital Ocean**, with **Firebase** managing real-time data synchronization and **PostgreSQL** storing structured agricultural data.

VI. RESULTS AND DISCUSSION

The AgriVerseAI system was rigorously tested with a group of 50 Kannada-speaking farmers from the districts of Mandya, Tumakuru, and Dharwad. The evaluation focused on several critical metrics to assess the system's effectiveness and usability

Evaluation Metric	Score (Out of 5)	Remarks
Voice Recognition Accuracy	4.6	Reliable STT performance in both Kannada and English across dialects.
Language Appropriateness	4.8	Effectively handled regional Kannada variations and provided clear replies.
Ease of Use	4.7	Voice-first design proved accessible to low-

		literacy users.
Helpfulness of Advice	4.9	Responses were actionable, accurate, and highly relevant.
Overall Satisfaction	4.8	Strong user acceptance and positive engagement from farmers.

Participant feedback emphasized the importance of Kannada language support and voice interaction, which significantly lowered barriers to technology adoption among rural farmers. These results validate AgriVerseAI's potential as an inclusive tool for agricultural knowledge dissemination.

VII. LIMITATIONS

Despite its strengths, AgriVerseAI faces several limitations that impact its current performance. The system's speech recognition accuracy is affected by variations in Kannada dialects across Karnataka, leading to occasional misunderstandings or misclassifications. Additionally, AgriVerseAI relies heavily on continuous internet connectivity to access cloud-based NLP, speech-to-text (STT), and text-to-speech (TTS) services, which may not be consistently available in remote rural areas. Although offline support is a future goal, it is not yet fully implemented. Furthermore, real-time market price updates depend on third-party APIs such as Agmark net, whose data availability and reliability can vary, potentially impacting advisory quality. Addressing these limitations through dialect adaptation, offline capability development, and enhanced data integration remains critical for improving system robustness and scalability.

VIII. FUTURE IMPLIMENTATIONS

Future enhancements of AgriVerseAI aim to further empower Karnataka's farmers by expanding its technological capabilities and accessibility. One key development is the integration of image-based disease detection using convolutional neural networks (CNN), enabling farmers to capture photos of affected crops for accurate and rapid diagnosis. Additionally, introducing Interactive Voice Response (IVR) support will facilitate offline phone access, allowing users without smartphones or stable internet to engage with the assistant via basic calls. Another major goal is to train a dedicated Kannada language large language model (LLM) using open-source frameworks like Llama or Mistral, enhancing contextual understanding and response generation tailored to regional expressions. Lastly, expanding dialectal coverage and voice adaptation will personalize interactions across Karnataka's diverse linguistic landscape, improving usability and trust.

IX. CONCLUSITION

AgriVerseAI represents a significant advancement in inclusive digital agriculture by providing Karnataka's farmers with a bilingual, voice-first assistant that bridges language and literacy barriers. Through seamless interaction in Kannada and English, it delivers localized, reliable, and timely agricultural information covering key farming domains. This tailored solution enhances farmers' decision-making capabilities and access to government schemes, market trends, and crop management knowledge without requiring literacy or technical expertise. By empowering marginalized rural communities with context-aware AI-driven guidance, AgriVerseAI exemplifies the growing potential of smart agriculture tools to foster equitable rural development. As India's agriculture sector evolves, such intelligent assistants will be critical for ensuring technological benefits reach all farmers, promoting sustainability, productivity, and inclusive growth

X. REFERENCES

[1] Abade, A., Ferreira, P. A., & de Barros Vidal, F. (2021). Plant diseases recognition on images using convolutional neural networks: A systematic review. Computers and Electronics in Agriculture, 185, 106125. https://doi.org/10.1016/j.compag.2021.106125

[2] Abdullah, H. M., Mohana, N. T., Khan, B. M., Ahmed, S. M., Hossain, M., Islam, K. S., ... & Ahamed, T. (2023). Present and future scopes and challenges of plant pest and disease (P&D) monitoring: Remote sensing, image processing, and artificial intelligence perspectives. Remote Sensing Applications: Society and Environment, 32, 100996. https://doi.org/10.1016/j.rsase.2023.100996

[3] Aceto, G., Persico, V., & Pascale, A. (2020). Industry 4.0 and health: Internet of things, big data, and cloud computing for healthcare 4.0. Journal of Industrial Information Integration, 18, 100129. https://doi.org/10.1016/j.jii.2020.100129

[4] Ahad, M. T., Li, Y., Song, B., & Bhuiyan, T. (2023). Comparison of CNN-based deep learning architectures for rice diseases classification. Artificial Intelligence in Agriculture, 9, 22–35. https://doi.org/10.1016/j.aiia.2023.03.003

[5] Ali, F., Razzaq, A., Tariq, W., Hameed, A., Rehman, A., Razzaq, K., ... & Ondrasek, G. (2024). Spectral Intelligence: AI-Driven Hyperspectral Imaging for Agricultural and Ecosystem Monitoring. [Journal Name] (awaiting full citation details).

[6] Bayer, P. E., & Edwards, D. (2021). The evolution of agricultural practices with machine learning: Shifting from isolated systems to integrated solutions. Plant Biotechnology Journal, 19(4), 648–650.https://doi.org/10.1111/pbi.13481

[7] Aashu, Rajwar, K., Pant, M., & Deep, K. (2024). Recent innovations and future directions for machine learning in farming systems. arXiv preprint, arXiv:2405.17465. https://doi.org/10.48550/arXiv.2405.17465

[8] Reddy, G. S., Reddy, M., Joshi, A., Chaitanya, K., & Sahithi. (2025). Adoption of machine learning in agriculture: Practical challenges and research scope. Journal of Experimental Agricultural Studies, 47(1), 43–57.

[9] Modi, D., Sutagundar, A. V., Yalavigi, V., & Aravatagimath, A. (2021). Building intelligent crop suggestion models with machine learning. In Proceedings of ISCON 2021 Conference on Information Systems and Computer Networks.